

Higher-order clustering for 3D point clouds

Exact density hierarchies and geometric tokens for 3D learning

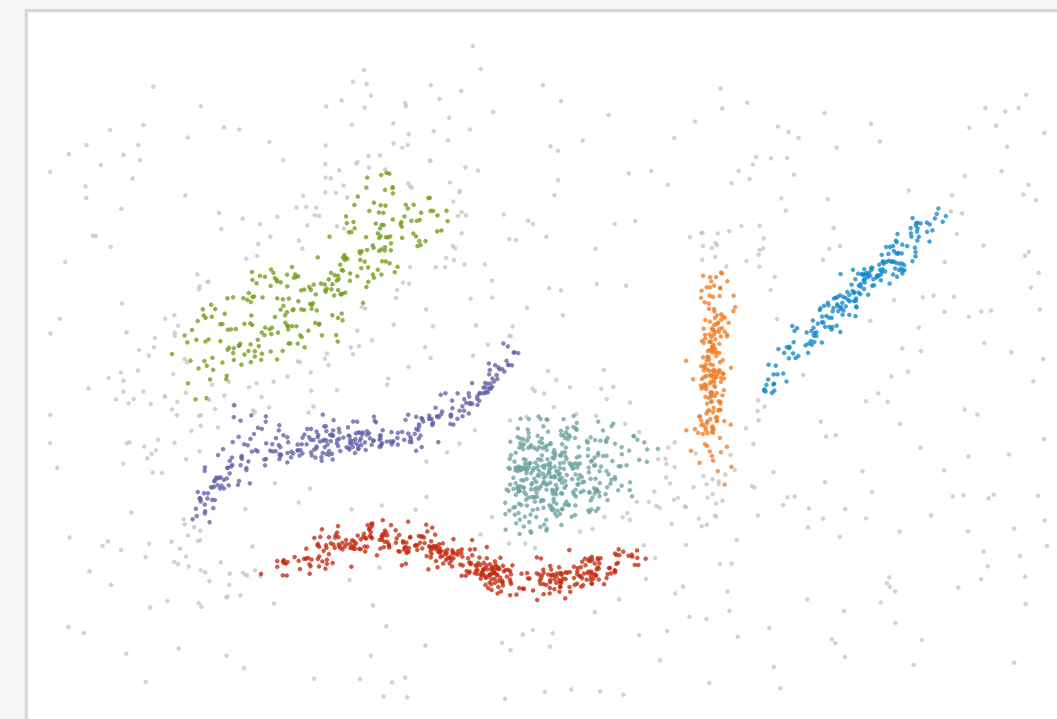
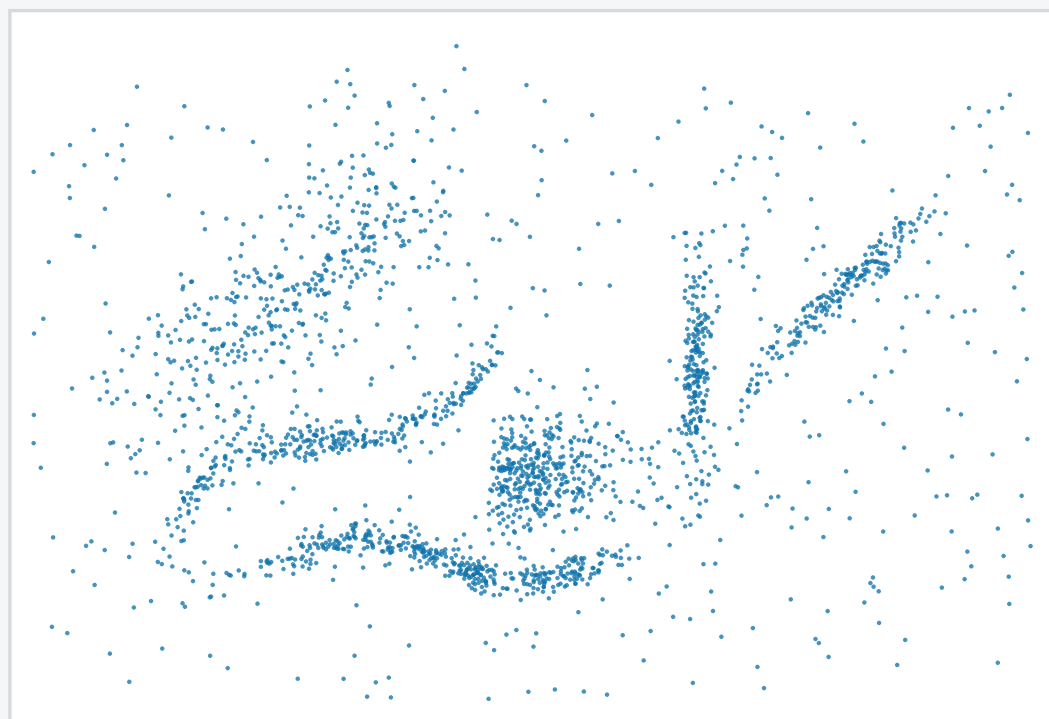
Louis HAUSEUX Konstantin AVRACHENKOV Josiane ZERUBIA
Inria, Université Côte d'Azur · NEO & AYANA teams · 3IA-DS4H

1 Clustering

Clustering groups similar points into *clusters*. Here, similarity is measured by Euclidean distance.

Unlabelled points

HDBSCAN [3]



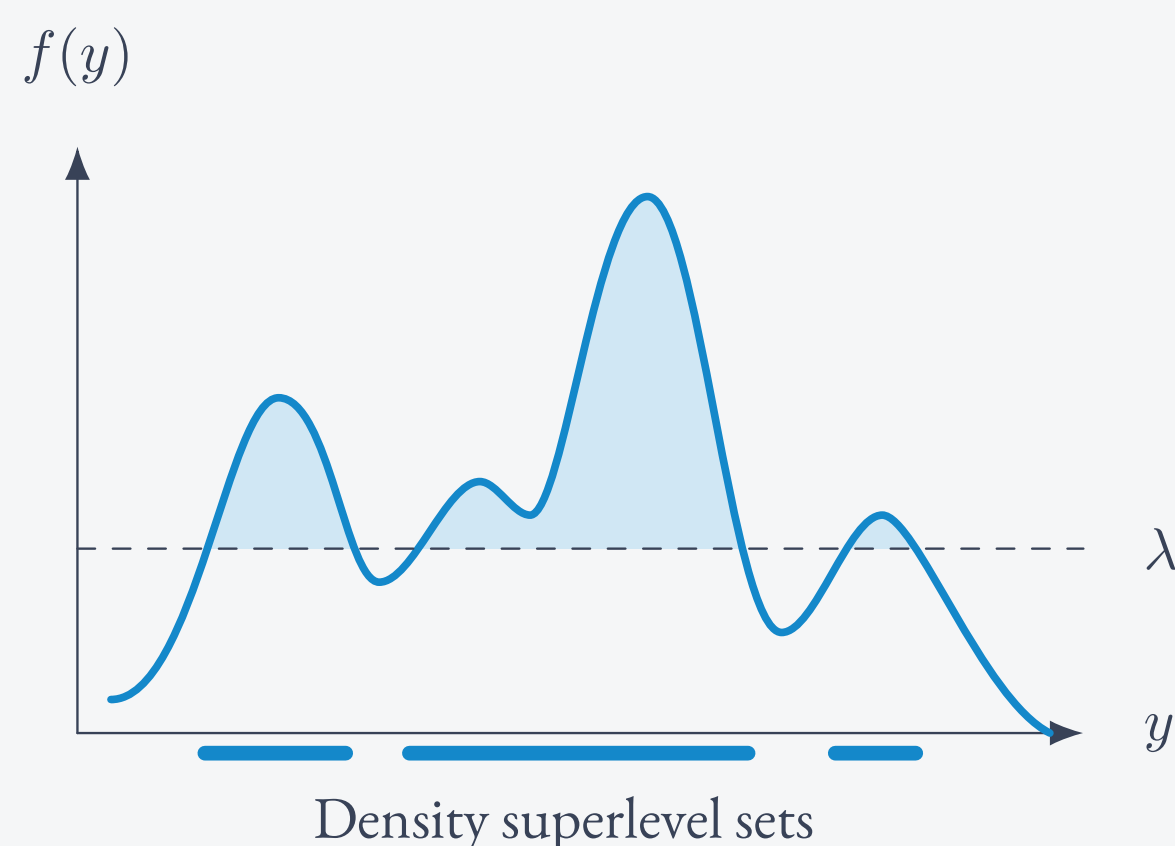
Clusters in colour; unassigned points in grey [3].

2 HARTIGAN's model and hierarchy

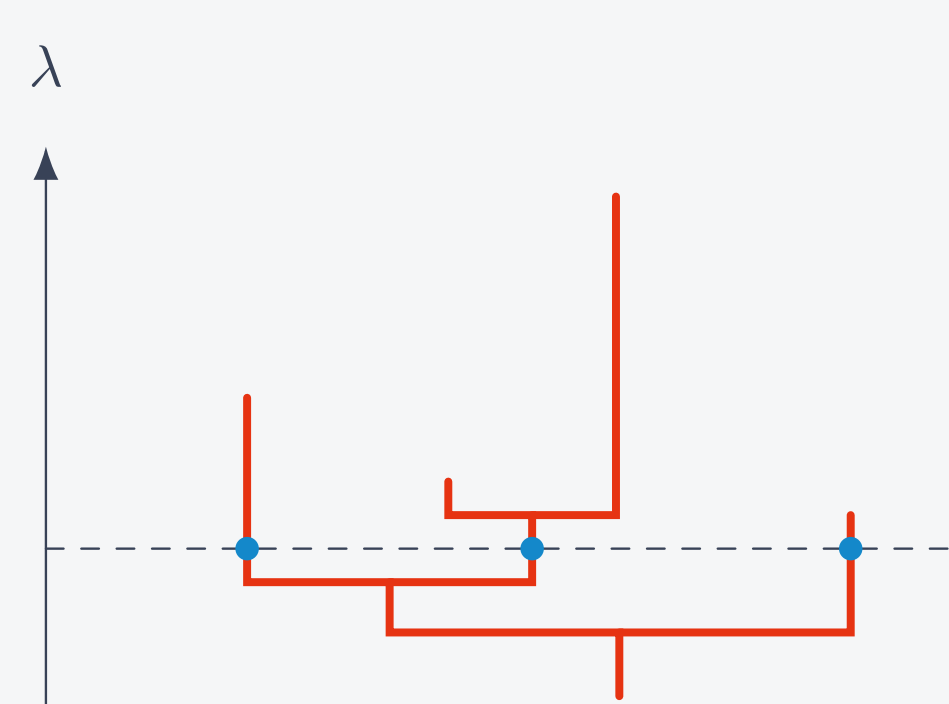
For a density f , clusters are the **connected components** of its superlevel sets [1].

$$E_\lambda = \{y \in \mathbb{R}^d : f(y) \geq \lambda\}$$

Lowering λ makes components grow and merge: a **hierarchy**.



Density superlevel sets



The same clusters in the tree

We seek to **estimate the unknown density f** . A natural, non-parametric choice is the K -nearest-neighbour estimator [2]:

$$\hat{f} \leftarrow \hat{f}_{K\text{-NN}}$$

$$\hat{f}_{K\text{-NN}}(y) = \frac{K}{n\omega_d r_K(y)^d}$$

$r_K(y)$: distance to the K -th nearest sample point.

n : number of points; ω_d : unit-ball volume.

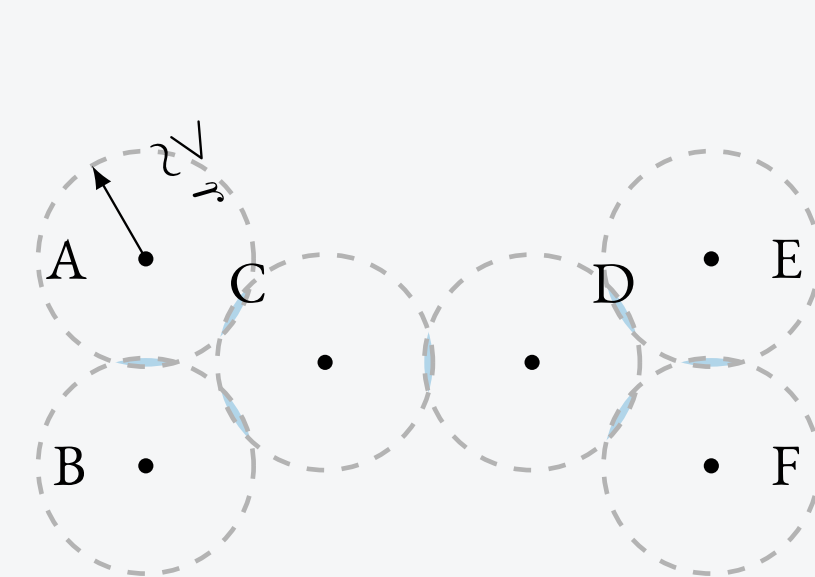
Change of variable: density level to neighbour radius.

$$\hat{f}_{K\text{-NN}}(y) \geq \lambda \iff r_K(y) \leq (K/(n\omega_d\lambda))^{1/d} =: r(\lambda)$$

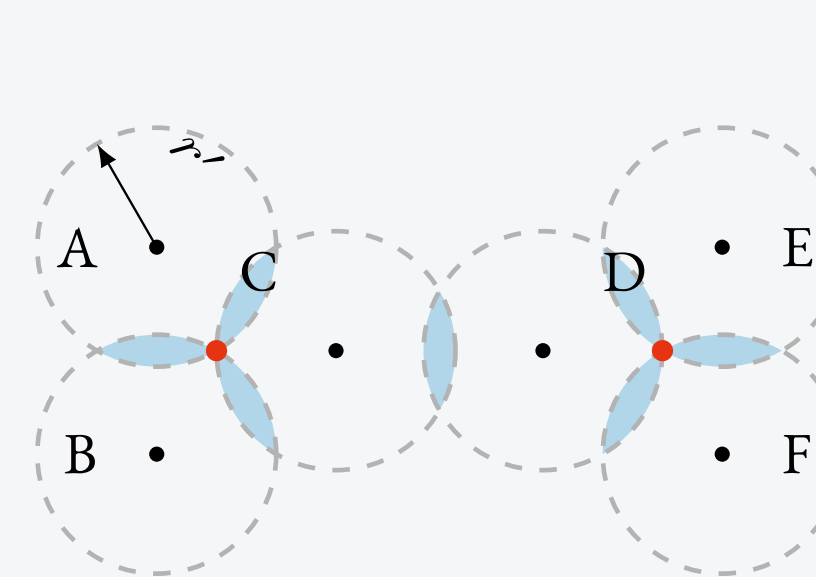
3 Same six points, different hierarchies

For $K = 2$, a density superlevel set is the region covered by **at least two balls**:

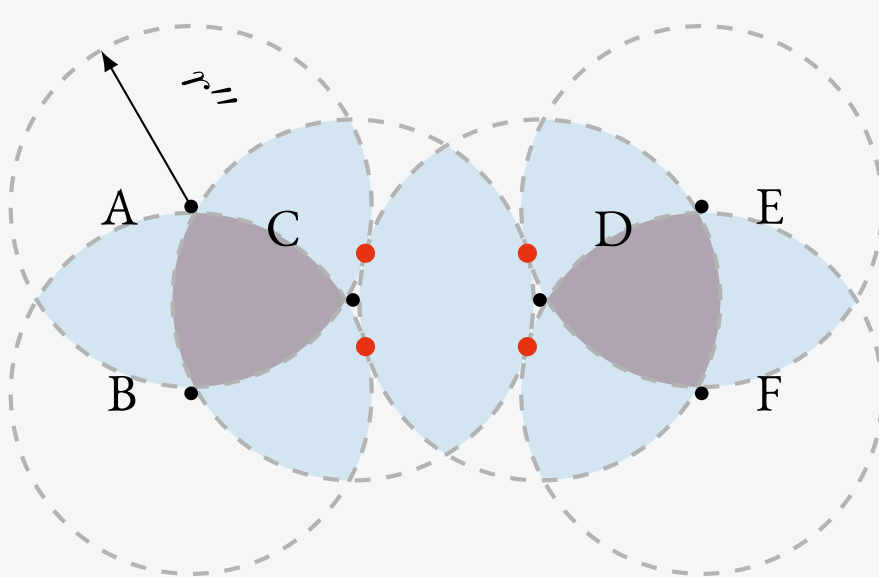
$$\{\hat{f}_{2\text{-NN}} \geq \lambda\} = \bigcup_{i < j} (B(x_i, r(\lambda)) \cap B(x_j, r(\lambda)))$$



r : 7 overlaps



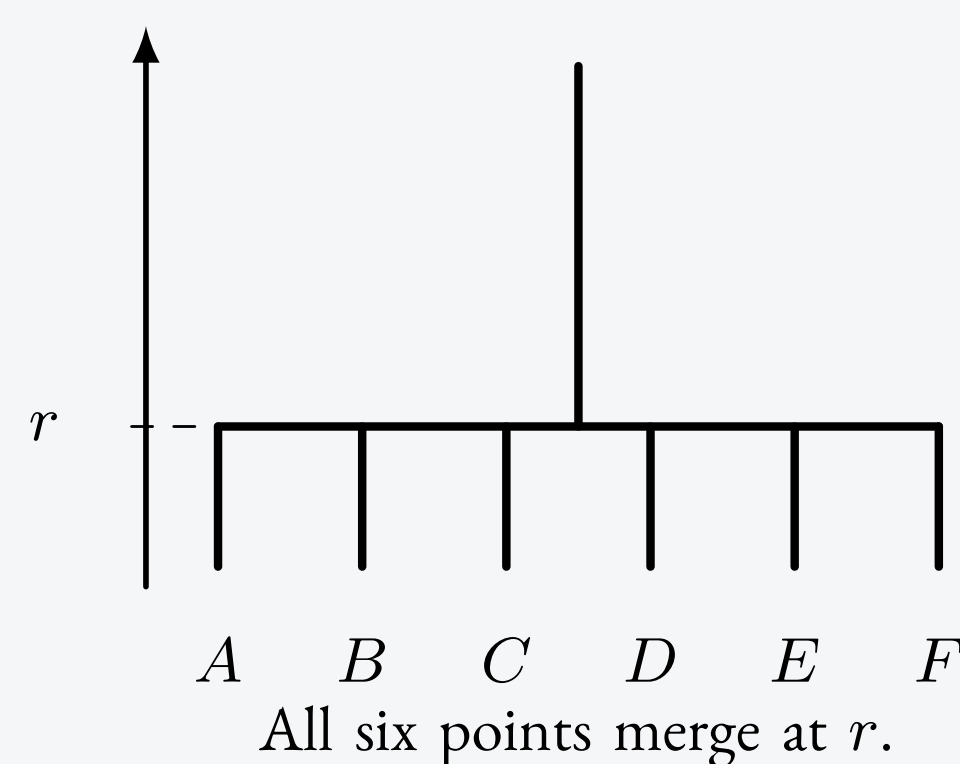
r' : 3 components



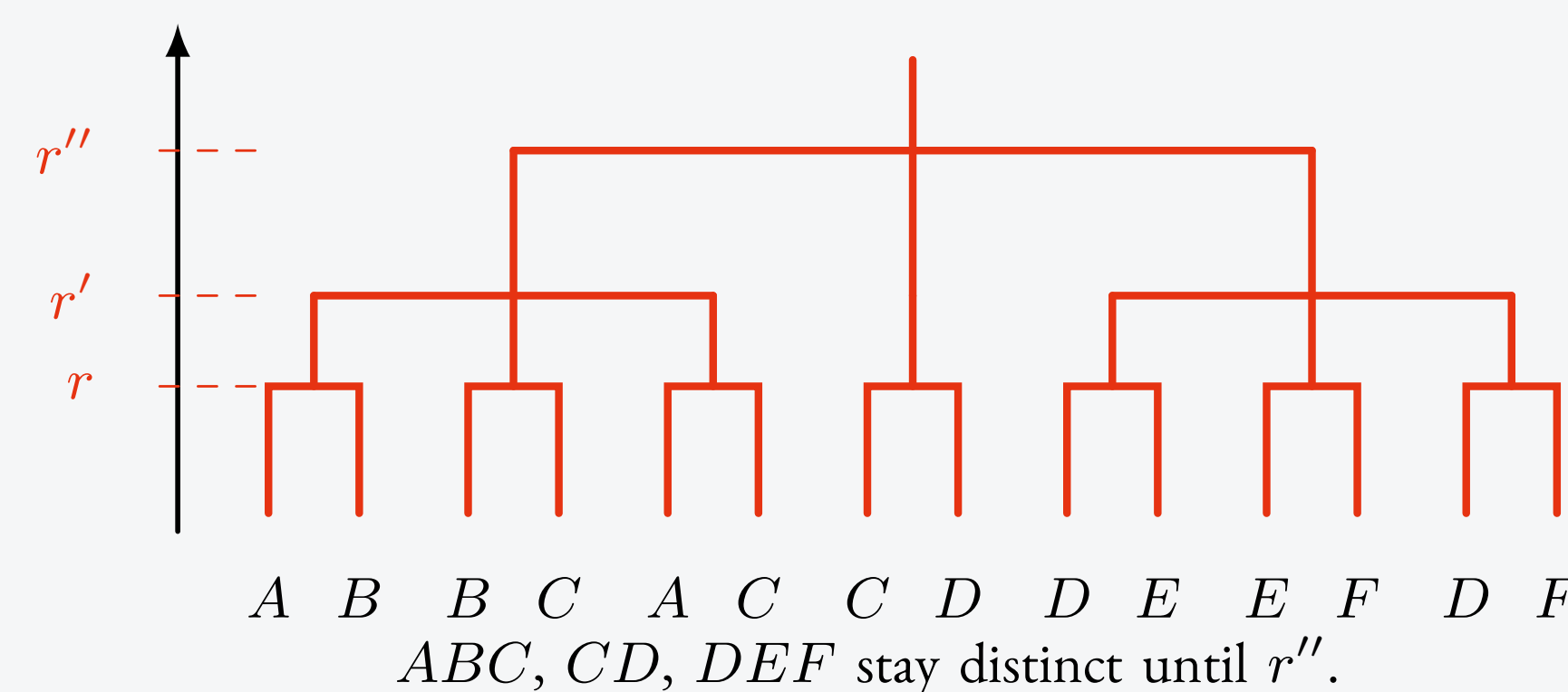
r'' : 1 component

DBSCAN / Robust SL

2-NN HARTIGAN



All six points merge at r .



ABC, CD, DEF stay distinct until r'' .

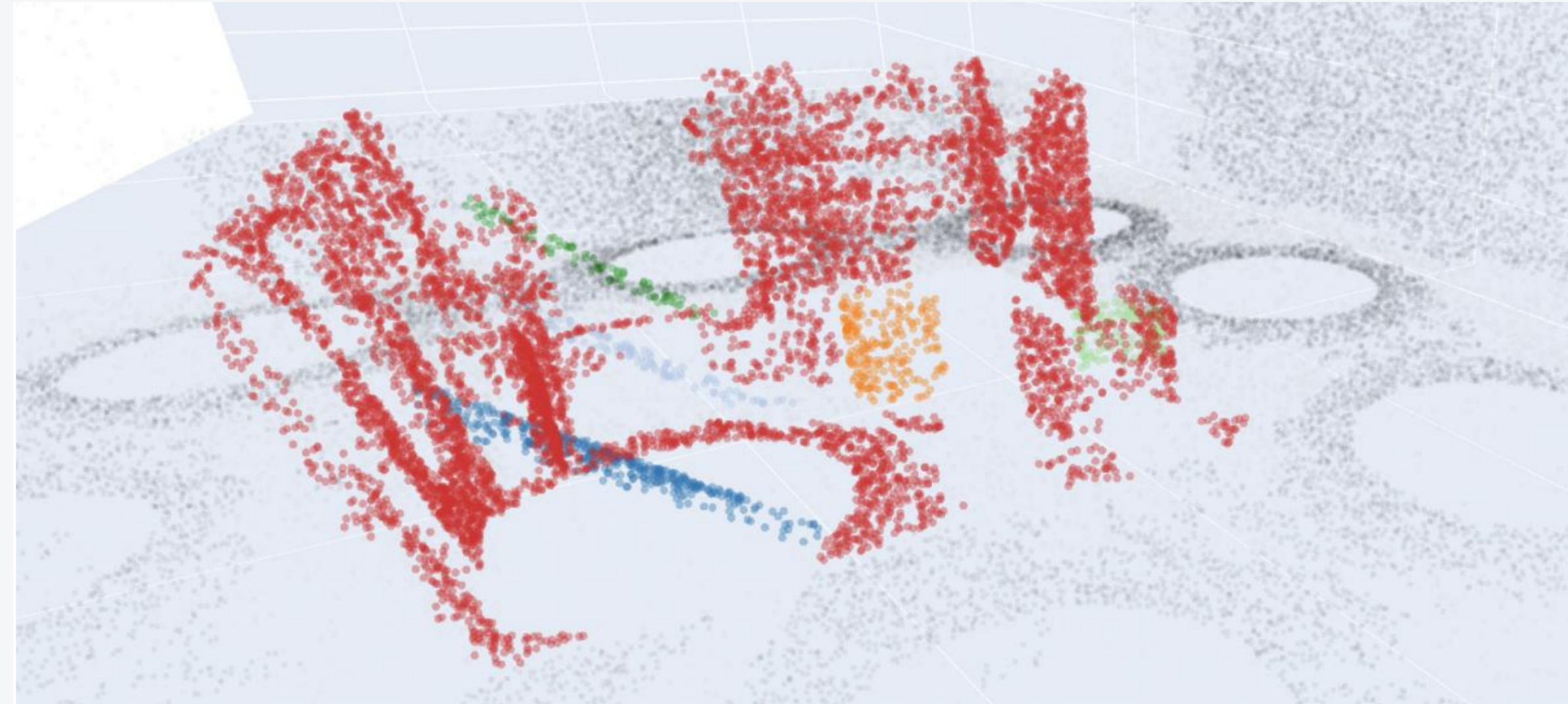
HGP-Clusterer connects edges through triangles to recover the exact hierarchy [4,5].

7 References

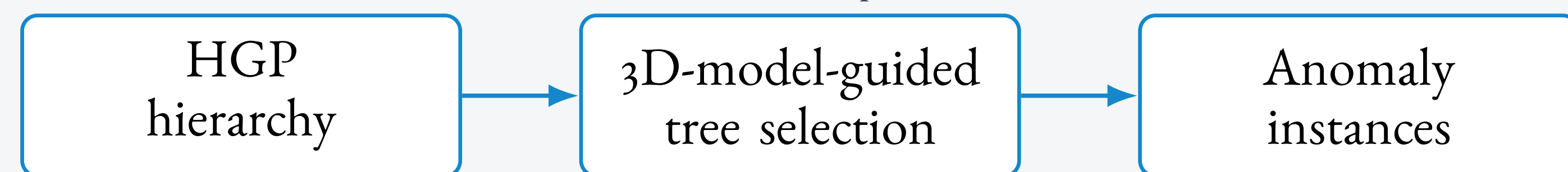
- [1] J. A. Hartigan. *Consistency of single linkage for high-density clusters*. JASA, 76:388–394, 1981.
- [2] G. Biau and L. Devroye. *Lectures on the Nearest Neighbor Method*. Springer, 2015.
- [3] R. J. G. B. Campello, D. Moulavi and J. Sander. *Density-Based Clustering Based on Hierarchical Density Estimates*. PAKDD, pp. 160–172, 2013.
- [4] L. Hauseux, K. Avrachenkov and J. Zerubia. *Generalization of single-linkage with higher-order interactions*. Applied Network Science, 11, article 9, 2026.

4 LiDAR: detect unexpected objects

Objects present in a scan, absent from a reference 3D model. With Marie ASPRO (Inria Startup Studio).



Synthetic benchmark: 2 million points. Red: model-consistent; other colours: anomalies; grey: background. © Naval Group [6].

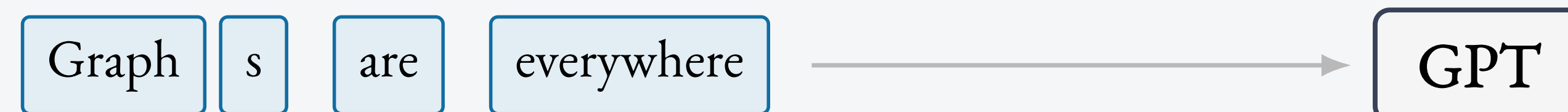


Geometric priors select branches. No training required.

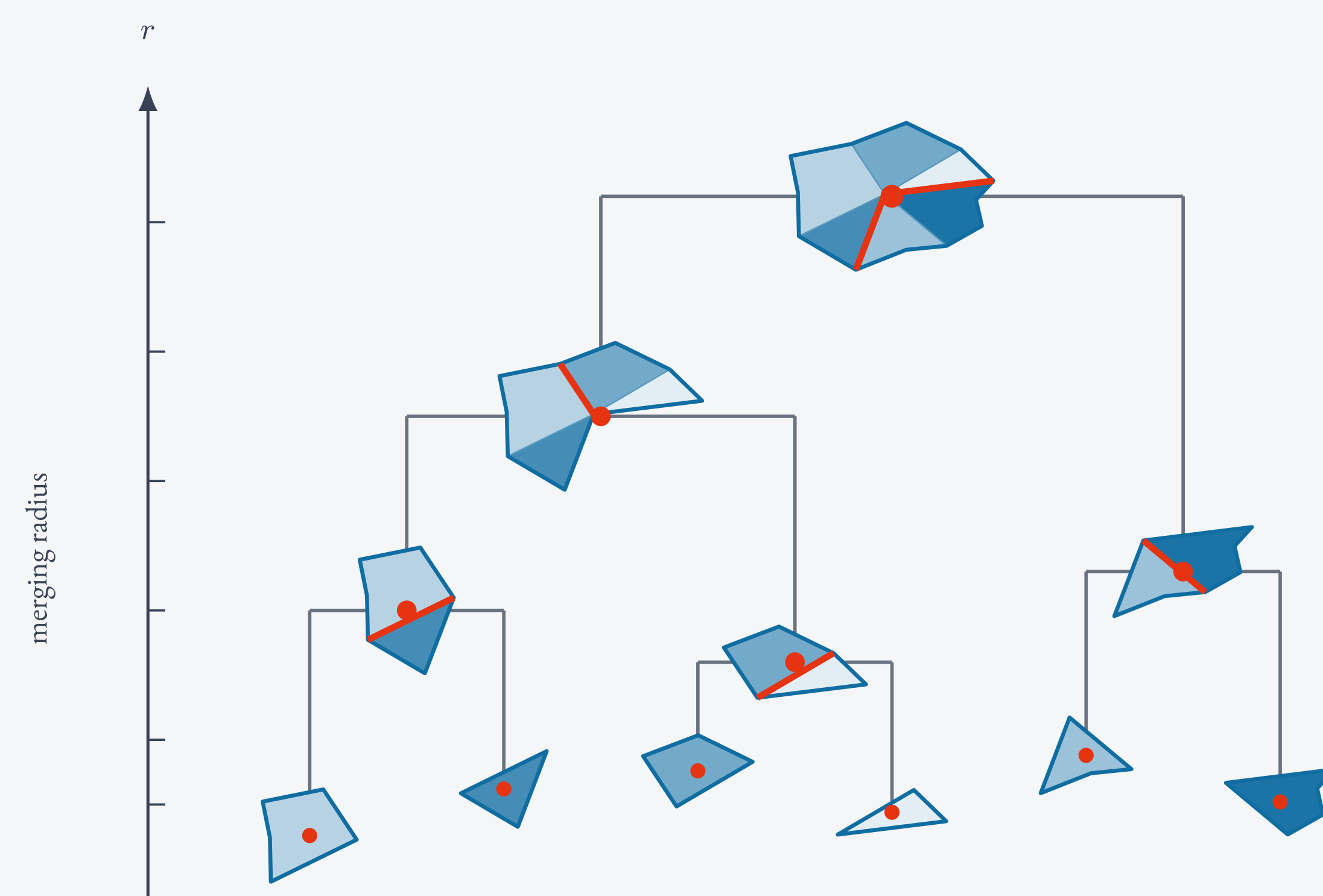
5 Perspective: geometry-guided 3D learning

GPT splits text into **tokens**, often parts of words [7].

It learns how these tokens fit together.



In 3D, our idea is to use **geometric pieces as tokens**, rather than individual points.



Research hypothesis: guide a 3D foundation model with polyhedral tokens and multiscale context from their merging hierarchy.

6 Inria Startup Studio: Percolia



Louis HAUSEUX & Alban HAUSEUX
Entering Inria Startup Studio.
Geometry-aware tools for 3D perception.



LinkedIn